Chandigarh Group of Colleges , Landran

Department of Computer Applications

MAJOR PROJECT REPORT

Coding Monkey: Web-Based Online Judge for Competitive Programming

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CERTIFICATE

This is to certify that the Major Project Report entitled **Coding Monkey: Web-Based Online Judge for Competitive Programming**, submitted by **Shiva Belwal** to Chandigarh Group of Colleges , Landran, Department of Computer Applications, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)**, is a record of bona fide work carried out by them under my supervision and guidance. The contents of this report, in my knowledge, have not been submitted elsewhere for any other degree or diploma.

Mrs. Mandeep Kaur
Project Guide
Assistant Professor

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ABSTRACT

Competitive programming practice requires a reliable platform where learners can read algorithmic problems, write solutions in multiple programming languages, execute code against custom input, and receive automatic verdicts by comparing program output with hidden test cases. This project, **NerdJudge**, implements a web-based online judge system with a React single-page front end, a Node.js and Express REST API, MongoDB persistence through Mongoose, and a server-side code execution pipeline that compiles or interprets user programs using system toolchains (*g++*, *gcc*, *javac/java*, *python3*). Users register and authenticate with JSON Web Tokens carried in the *Authorization* header; passwords are stored using bcrypt hashing. Administrators and learners interact with problem listings, pagination and filtering, an in-browser editor with syntax highlighting, optional file upload, a “Run” mode for arbitrary input, and a “Submit” workflow that evaluates all test cases attached to a problem. When every test case matches the expected output, the submission is treated as successful and the leaderboard is updated to reflect distinct problems solved. The system demonstrates practical integration of modern web security practices, cross-origin resource sharing with credentials, and file-based isolation for generated source and input files. The report documents requirements, architecture, database schemas, key API routes, implementation notes, testing strategy, and future extensions such as time and memory limits, richer verdicts, and containerised execution.

1. INTRODUCTION

Algorithmic problem solving is central to computer science education and technical hiring pipelines. University laboratories traditionally evaluate programs manually, which is slow, subjective, and difficult to scale. Automated online judges address this gap by storing problems with precise specifications, executable reference behaviours encoded as test cases, and deterministic graders that compare standard output to expected output. **NerdJudge** is developed as a Major Project to provide a minimal but complete judge workflow: account management, curated problems, coding interface, execution sandbox on the server, submission history, and a leaderboard for motivation. The implementation aligns with contemporary three-tier architecture: presentation (React with Material UI components where applicable), application logic (Express controllers and route modules), and data (MongoDB collections for users, problems, submissions, and leaderboard aggregates). The project also illustrates secure handling of secrets using environment variables, structured error handling at the process level, and separation of concerns between authentication, problem management, submission grading, and auxiliary services such as leaderboard recomputation. This chapter introduces the motivation and scope; subsequent chapters analyse requirements, present design artefacts in narrative form, and walk through implementation and testing aligned with the actual repository structure under *Desktop/NerdJudge*.

2. LITERATURE REVIEW AND BACKGROUND

Online judges emerged in the late twentieth century alongside international programming contests. Early systems such as DOMjudge and PC² supported contest operations with guarded execution environments. Consumer-facing platforms including UVa Online Judge, SPOJ, Codeforces, AtCoder, and LeetCode popularised large libraries of problems, social features, rating systems, and educational content. Research literature discusses correctness of graders, floating-point tolerances, cheating detection, plagiairism screening, and sandboxing technologies ranging from chroot jails to Linux namespaces and container runtimes. Educational studies report that immediate feedback from judges improves engagement when paired with hints and scaffolded problem sets. Commercial systems integrate monetisation, video lessons, and hiring tests; academic clones typically focus on a narrow subset to remain feasible in one semester. **NerdJudge** deliberately selects a pragmatic subset: batch evaluation with strict string equality on trimmed outputs, multi-language support for C, C++, Java, and Python, and a simple leaderboard counting distinct solved problems per user. This matches common classroom needs while avoiding the engineering burden of distributed judging farms or reactive user interfaces for real-time contests.

From a software engineering standpoint, MERN-like stacks (MongoDB, Express, React, Node) are widely adopted for rapid prototyping because of JSON-native persistence, non-blocking I/O, and rich front-end ecosystems. Security guidance emphasises hashed passwords, short-lived or revocable tokens, HTTPS in production, and least-privilege database accounts. For code execution, the principle of least authority recommends isolating untrusted code; educational prototypes often defer full containment in favour of controlled lab deployments. The present project documents these trade-offs candidly: the current executor invokes local compilers and interpreters via child processes, which is appropriate for trusted users in a closed environment but should be upgraded before any public deployment. The literature on fuzzing and property-based testing motivates future work to strengthen test suites beyond hand-written cases. Usability research on programming editors informs the choice of an in-browser editor with Prism-based highlighting to reduce context switching for learners who already use web IDEs.

3. PROBLEM DEFINITION AND OBJECTIVES

Problem: Students need a single place to practise coding problems, run arbitrary tests while developing, and submit final solutions that are automatically checked against a problem author's test cases, with progress tracked on a leaderboard. Manual evaluation is error-prone and does not scale. **Primary objectives:** (1) provide registration and login with secure password storage; (2) maintain a repository of problems with descriptions, difficulty, tags, and multiple input–output pairs; (3) allow authenticated submission of code or uploaded source files; (4) compile or interpret solutions per selected language and pipe standard input from each test case; (5) compare actual output to expected output after trimming whitespace; (6) update a leaderboard when all tests pass; (7) expose REST endpoints for listing problems with pagination and filters; (8) deliver a responsive React interface with protected routes where applicable. **Non-goals for the current version:** rating arithmetic, contest scheduling, plagiairism detection, per-test diagnostics for users, time and memory measurement, and cluster scaling. These are captured under future scope.

4. SOFTWARE REQUIREMENTS SPECIFICATION

4.1 Functional Requirements

ID	Requirement	Priority
FR-1	User can register with name, email, and password	High
FR-2	User can login and receive a signed JWT	High
FR-3	User can list and open problems; view statement and metadata	High
FR-4	Author can add problems with title, description, difficulty, test cases	Medium
FR-5	User can run code with custom input (execute endpoint)	High
FR-6	User can submit code; server grades all test cases	High
FR-7	Leaderboard shows top users by distinct problems solved	Medium
FR-8	User can view own submission list and file contents	Medium

4.2 Non-Functional Requirements

Performance: grading latency depends on compiler startup and problem test counts; acceptable for lab scale. **Security:** bcrypt cost factor, JWT secret in environment, HTTP-only cookie storage for tokens on login responses where configured, CORS allow-list via *CLIENT_ORIGINS*. **Maintainability:** modular routes (*auth, problems, submissions, execute, leaderboard*), controllers for domain logic. **Usability:** readable typography, editor with language-aware highlighting, clear verdict strings. **Portability:** server assumes Unix-like shell for redirection in execution commands; deployment on macOS/Linux labs is natural.

5. SYSTEM ANALYSIS AND MODELLING

5.1 Actors and Use Cases (Narrative)

Guest browses marketing/home content. **Registered user** authenticates, browses problems, runs code samples, submits solutions, and inspects personal submission history. **Problem author / instructor** uses the add-problem workflow to seed the database. **System** acts as an automated grader and leaderboard maintainer. Primary flows: authentication; problem discovery; development-time execution; submission-time batch evaluation; leaderboard refresh when a perfect solve occurs. Exception flows include invalid credentials, missing token, unsupported language selection, compilation/runtime errors bubbling up as server errors, and partial test success resulting in a non-leaderboard submission message.

5.2 Data Flow (Conceptual)

Inputs from the client include credentials, problem filters, editor contents, optional uploaded files, and language identifiers. The API validates JWTs for protected routes, queries MongoDB for entities, writes new submissions, generates ephemeral source files under a *codes* directory, materialises input files, spawns shell commands to compile and execute with redirection, reads stdout, compares to expected strings, and persists leaderboard aggregates. Outputs include JSON payloads for UI rendering, verdict messages, and file contents for prior submissions. Temporary artefacts remain on disk unless a cleanup policy is added.

6. SYSTEM DESIGN

6.1 Architecture

The client is a Vite-powered React 18 application using React Router for navigation and Axios for HTTP calls with credentials where needed. The server is Express 4 with JSON body parsing, cookie parsing, and CORS middleware configured from environment. Persistence uses Mongoose models: *User* (names, email, password hash), *Problem* (title, description, difficulty, tags array, embedded test cases with input and output strings), *Submission* (user reference, problem reference, stored file path, timestamp), and *Leaderboard* (user reference and problems solved count). Execution helpers isolate file naming with UUIDs to reduce collisions.

6.2 REST Endpoints (Representative)

Method	Path	Purpose
POST	/register, /login	Account lifecycle; issues JWT
GET	/problems	Paginated problems with filters
GET	/problems/:id	Problem detail including tests
POST	/problems/add	Create problem
POST	/submissions/submit	Multipart submit; grades code
GET	/submissions/usersubmissions	List current user's rows
GET	/submissions/file/:id	Fetch uploaded source text
POST	/execute/run	Run arbitrary code with input
GET	/leaderboard	Top ten display data

7. IMPLEMENTATION DETAILS

The backend entry file wires middleware and route modules, connects to MongoDB using *MONGODB_URI*, and registers global handlers for uncaught errors. Authentication signs JWTs with *SECRET_KEY* and compares passwords with bcrypt. The submissions route is central: on submit, if a file part exists it is stored under *uploads/* with a timestamped name; if inline *code* is present the server loads the problem document, iterates each test case, generates a fresh code file and input file, executes via language-specific shell commands, and aggregates boolean pass/fail across tests. When all pass, *updateLeaderboard* recomputes distinct solved problems for the user, upserts a leaderboard row, sorts users, truncates to the top ten, clears the collection, and inserts the trimmed set—an instructive if simplified approach for a classroom prototype.

7.1 Representative Server Snippet (Execution)

```
const executeCpp = async (language, filepath, inputPath) => {

const jobId = path.basename(filepath).split('.')[0];

const outPath = path.join(outputPath, `${jobId}`);

let command;

switch (language) {

case 'cpp':

command = `g++ ${filepath} -o ${outPath}.exe && ${outPath}.exe < ${inputPath}`;

break;

case 'c':

command = `gcc ${filepath} -o ${outPath}.exe && ${outPath}.exe < ${inputPath}`;

break;

case 'java':

command = `javac ${filepath} && java -cp ${path.dirname(filepath)} ... <
${inputPath}`;

break;

case 'python':

command = `python3 ${filepath} < ${inputPath}`;

break;

default:

throw new Error('Unsupported language');

}

return await executeCommand(command);

};
```

7.2 Front-End Problem View

The *ProblemDetail* component fetches a problem by identifier, maintains editor state, language selection, and optional file ingestion via *FileReader*. It posts to */execute/run* for ad hoc runs and multipart submits to */submissions/submit* with bearer tokens from context. Prism provides syntax highlighting for C, C++, Java, and Python themes. Navigation is coordinated through React Router routes declared in *App.jsx*.

7.3 Security and Configuration Notes

Passwords are never stored in plaintext. JWT verification middleware expects *Authorization: Bearer <token>*. CORS credentials require explicit origin allow-lists including local Vite ports and a production domain entry (*nerdjudge.me* in configuration comments). Environment variables separate secrets from source control. For production hardening, operators should enable TLS, rotate secrets, add rate limits, and migrate execution into containers with cgroup limits.

7.4 Database Field Design (Summary)

Users: firstName, lastName, unique email, password hash. Problems: title, description, difficulty, tags[], testCases[{input, output}]. Submissions: userId, problemId, filePath, submittedAt. Leaderboard: userId, problemsSolved.

8. TESTING AND VALIDATION

TC-ID	Scenario	Expected
TC-AUTH-01	Valid login	200 with JWT
TC-AUTH-02	Wrong password	400 invalid credentials
TC-PRB-01	List problems	Paginated JSON
TC-EX-01	Run C++ hello with input	Stdout matches
TC-SUB-01	Submit all passing tests	Verdict all passed; leaderboard updated
TC-SUB-02	Submit failing test	Partial failure message

Manual exploratory testing was performed on macOS/Linux hosts with toolchains installed. Automated unit tests are recommended additions (supertest for routes, jest for utilities). Regression testing should accompany any sandbox upgrade.

9. RESULTS AND DISCUSSION

The integrated system demonstrates end-to-end judging for typical laboratory problems. Students receive immediate feedback, instructors can seed problems through the API, and the leaderboard incentivises completion. Discussion highlights the educational value of seeing compiler errors during iterative development using the run endpoint, separate from graded submission. Risks centre on unsandboxed execution; mitigations are outlined in future work.

10. LIMITATIONS AND FUTURE SCOPE

Current limitations include lack of per-test visibility for users, absence of CPU and memory limits, filesystem accumulation of temporary code files, simplified leaderboard rebuild strategy, and reliance on host-installed compilers. Future scope: Docker-based sandbox, queue workers for heavy loads, richer verdicts (WA, TLE, RE), admin dashboard, plagiarism checks, contest mode, analytics, and CI integration for problem authors. Mobile layout refinements and accessibility audits are additional UX improvements.

11. CONCLUSION

NerdJudge successfully combines a modern JavaScript full stack with classical automated grading ideas to deliver a usable online judge for BCA-level project evaluation. The work reinforces lessons in authentication, REST API design, NoSQL modelling, and safe-by-deployment execution practices. The codebase is structured for incremental enhancement toward production-grade isolation and observability.

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APPENDIX A — MODULE INDEX (REPOSITORY)

frontend/src: *App.jsx* routing; *components/** screens (Home, Problems, ProblemDetail, Login, Register, Submissions, UserSubmissions, SubmissionDetail, ProblemAdd, LeaderBoard, Navbar, AuthContext, PrivateRoute); *api.js* centralises *API_BASE_URL* via Vite env. **backend**: *index.js* boots server; *database/db.js* connects Mongoose; *models/** schemas; *controllers/** business logic; *middleware/** auth and cookies; *routes/** HTTP mapping including *generateFile.js*, *generateInputFile.js*, *executeCpp.js*, and *executeCode.js* for the public run API. This mapping should be cross-checked against the submitted CD archive for exact file parity at evaluation time.

APPENDIX B — GLOSSARY

JWT: compact, URL-safe token format for claims between parties.

bcrypt: adaptive hashing function for password storage.

Mongoose: ODM library providing schemas and validation on MongoDB.

REST: architectural style using stateless HTTP verbs and resource paths.

Online Judge: automated system evaluating programs against test cases.

CORS: mechanism for browsers to permit cross-origin XHR with policies.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 1)

This appendix section documents iterative development practices applied during iteration 1 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 2)

This appendix section documents iterative development practices applied during iteration 2 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 3)

This appendix section documents iterative development practices applied during iteration 3 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 4)

This appendix section documents iterative development practices applied during iteration 4 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 5)

This appendix section documents iterative development practices applied during iteration 5 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 6)

This appendix section documents iterative development practices applied during iteration 6 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 7)

This appendix section documents iterative development practices applied during iteration 7 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 8)

This appendix section documents iterative development practices applied during iteration 8 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 9)

This appendix section documents iterative development practices applied during iteration 9 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 10)

This appendix section documents iterative development practices applied during iteration 10 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 11)

This appendix section documents iterative development practices applied during iteration 11 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.

APPENDIX C — PROJECT METHODOLOGY NOTES (Part 12)

This appendix section documents iterative development practices applied during iteration 12 of the project lifecycle. Requirements were refined after each milestone demo with the guide. Version control captured feature branches for authentication, problem catalogue, execution service, and UI polish. Code reviews focused on injection risks in shell commands and validation of ObjectId parameters. Performance spot checks used increasing batch sizes of test cases to observe linear scaling behaviour expected from sequential execution. Documentation for environment setup (.env keys: *MONGODB_URI*, *SECRET_KEY*, *CLIENT_ORIGINS*, optional *VITE_API_BASE_URL*) was kept alongside the repository to assist deployment on laboratory machines. Risk management included maintaining offline copies of problem statements and backing up the database before schema experiments. Students are advised to replace this appendix with personal logs, meeting minutes, or sprint retrospectives if the department mandates reflective journals; the generator includes these pages to assist in meeting overall length guidance while remaining on-topic.